

Firmware Analysis Report

Project: firmware_test.bin
Architecture: Unknown
File Size: 4504 bytes
SHA256: 6878fc100602672da53cd3d0b8f03ad448f826bfddb13fa3f6f85bccbc75cafd

Executive Summary

This report was automatically generated by the FAWS system. It contains the results of a multi-stage static, dynamic, and symbolic firmware analysis.

Analysis Stages

Tool: STRINGS

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools/strings.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0

Starting Strings Analysis on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin... Extracted 114 strings. Found 23 potential secrets! [!] IPv4_ADDRESS: OBIS_CODE:1.0.1.8.0.255 [!] PRIVATE_KEY_MARKER: -----BEGIN RSA PRIVATE KEY----- [!] FILE_PATH: MQTT_BROKER_URL=mqtt://broker.hivemq.com:1883 [!] FILE_PATH: MQTT_BROKER_TLS_URL=mqtts://iot.eclipse.org:8883 [!] IPV4_ADDRESS: OTA_UPDATE_URL=http://192.168.1.1/update.php [!] URL: OTA_UPDATE_URL=http://192.168.1.1/update.php [!] FILE_PATH: OTA_UPDATE_URL=http://192.168.1.1/update.php [!] URL: CLOUD_API_URL=https://api.firmware-update.example.com/v1/ota [!] FILE_PATH: CLOUD_API_URL=https://api.firmware-update.example.com/v1/ota [!] IPV4_ADDRESS: 192.168.1.1 Scan complete.

Tool: BINWALK

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools/binwalk.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0

Starting Python Binwalk Engine on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin... Found 3 potential secrets! [!] Offset 0x0: ELF 32-bit LSB executable [!] Offset 0x9c7: Squashfs filesystem [!] Offset 0x9e6: gzip compressed data Scan complete.

Tool: CUTTER

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools/cutter.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0

Starting Cutter (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin... Found 1 potential secrets! [!]
Architecture: ELF executable detected Scan complete.

Tool: GHIDRA

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools/ghidra.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0

Starting Ghidra (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin... No interesting RE artifacts found. Scan complete.

Tool: TRUFFLEHOG

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools/trufflehog.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0

Starting Secret Scan on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin... Found 5 potential secrets! [!]
GENERIC_SECRET at line 3: TOKEN=ghp_FAKEToken1234567890abcdefFAKE [!] GENERIC_SECRET at line 3: password='Sup3rSecretPass [!] GENERIC_SECRET at line 3: PASSWORD=FactoryReset [!]
GENERIC_SECRET at line 13: TOKEN=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJkZXZpY2UtMDAxIiwicm9sZSI6ImFkbWwluIiwiaWF0IjoxNzAwMDAwMDAwfQ.RkFLRVNJR05BVFVSUJZVEVTTk9UUkVBTA [!]
GENERIC_SECRET at line 13: PASSWORD=WifiPass2024 Scan complete.

Tool: ENTROPY

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools/entropy.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0

Starting Entropy Analysis on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin... Overall Shannon Entropy: 6.2502 Entropy is within normal range for executable code. Scan complete.

Tool: WIRESHARK

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools\wireshark.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0

Starting Wireshark (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin... Found 1 potential secrets! [!]
Network: No PCAP signature, checking for HTTP strings... Scan complete.

Tool: AFL++

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools\afl.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0

Starting AFL++ (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin... Found 1 potential secrets! [!]
Fuzzing: Binary size suitable for efficient fuzzing Scan complete.

Tool: ANGR

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools\angr.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0

Starting angr (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin... Found 1 potential secrets! [!]
Symbolic: High number of branch instructions (144) indicates complex control flow suitable for symbolic execution Scan complete.

Tool: SCORECARD

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools\scorecard.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0

Starting Risk Scorecard Generation on project 17... Found 1 potential secrets! [!]
RiskScore: Calculated Aggregate Risk Score: 0/100 based on 22 findings Scan complete.

Tool: PDF_REPORT

Status: RUNNING | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\venv\Scripts\python.exe" tools\pdf_report.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\17_firmware_test.bin" --project "17" --run-id 0